

MOCK TEST: BIOMECHANICS AND SPORTS

PE Class 12 | Chapter 8 | Physics of Excellence

MAX MARKS: 50 TIME: 1 HOUR BOARD PATTERN

SECTION A: Very Short Answer (1m each)

QUESTION 1 • [1 MARK]

What is biomechanics in the context of sports?

QUESTION 2 • [1 MARK]

Name one principle of Newton's laws applied to sports, with an example.

QUESTION 3 • [1 MARK]

Define "center of gravity" and its importance in balance during sports.

QUESTION 4 • [1 MARK]

List one type of lever found in the human body during a bicep curl.

QUESTION 5 • [1 MARK]

What is the role of friction in sports like running?

SECTION B: Short Answer (3m each)

QUESTION 6 • [3 MARKS]

Explain the application of Newton's third law in sports. Provide one example from a game.

QUESTION 7 • [3 MARKS]

Discuss how projectile motion is used in sports like javelin throw. Mention factors affecting trajectory.

QUESTION 8 • [3 MARKS]

Describe the types of levers in the body and their mechanical advantages. Give examples.

QUESTION 9 • [3 MARKS]

How does the center of mass affect performance in gymnastics? Providing one maintenance strategy.

QUESTION 10 • [3 MARKS]

Outline the principles of force application in weightlifting. Mention force types.

SECTION C: Long Answer (5m each)

QUESTION 11 • [5 MARKS]

Elaborate on the principles of biomechanics and Newton's laws in improving performance.

QUESTION 12 • [5 MARKS]

Analyze the role of levers and force in movement. Provide class-wise examples and efficiencies.

QUESTION 13 • [5 MARKS]

Explain linear and angular motion. Discuss torque and momentum in swimming and discus throw.

ANSWER KEY & REVISION POINTS

A1: Study of mechanical laws relating to the movement or structure of living organisms.

A2: 2nd Law (Acceleration). Kicking a ball: Force applied determines the ball's speed.

A3: Point where the weight of a body is balanced. Low COG increases stability.

A4: Third-class lever (Effort between Fulcrum and Load).

A5: Provides traction for grip and propulsion; essential for starting/stopping sprints.

Summary (B & C): Projectile flight is governed by angle (45°) and velocity. Levers in the body (1st, 2nd, 3rd) translate muscle contraction into movement. 3rd class levers optimize speed over force. Angular motion is rotation around an axis, while torque is the rotational force used in shots and spins. Linear motion covers straight sprinting.

Mock Test Chapter 8 By Mindmaster Physical Education